

Multi-Role VR Training System for Film Production: Enhancing Collaboration with MetaCrew

Zheng Wei

The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, China
zwei302@connect.hkust-gz.edu.cn

Shan Jin

The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, China
sjin752@connect.hkust-gz.edu.cn

Wai Tong

Texas A&M University
College Station, United States of
America
wtong@tamu.edu

David Kei Man Yip

The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, China
daveyip@hkust-gz.edu.cn

Pan Hui*

The Hong Kong University of Science
and Technology (Guangzhou)
Guangzhou, China
panhui@hkust-gz.edu.cn

Xian Xu*

The Hong Kong University of Science
and Technology
Hong Kong SAR
xianxu@ust.hk

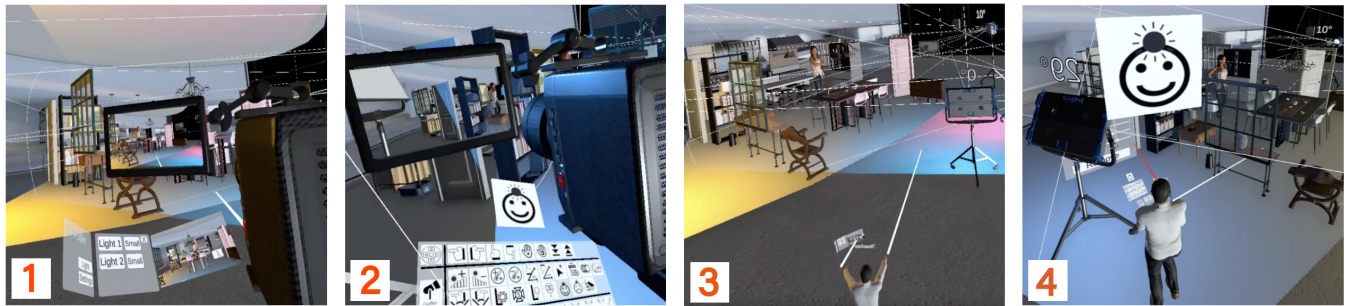


Figure 1: Presents a VR Collaborative Learning System for film production, showcasing (1 & 3) perspectives without, and (2 & 4) with a film operation GUI, from first and third-person views.

ABSTRACT

Film production within studios has become an industry standard in mature professional filmmaking. However, most training institutions cannot afford the expensive film production equipment essential for providing beginners with collaborative training alongside directors, cinematographers, cameramen, and other film professionals. To address these challenges, we first propose the Film Production Virtual Reality Collaborative Learning (VRCL) system – MetaCrew. We then developed an interface featuring visual symbols for film operations to facilitate smooth communication and learning among beginners using professional terminology in film production. In an experiment with 24 participants, we further explored the impact of the film operation graphical user interface (GUI) design on social interaction factors within our system. Results show that our system and GUI enhance social interaction and

learning outcomes among learners. Moreover, our system presents a smaller financial burden and requires less physical space.

CCS CONCEPTS

• Human-centered computing → Virtual reality; User studies.

KEYWORDS

Film Production Teaching, Collaborative Learning, Social Interaction, GUI, Virtual Reality

ACM Reference Format:

Zheng Wei, Shan Jin, Wai Tong, David Kei Man Yip, Pan Hui, and Xian Xu. 2024. Multi-Role VR Training System for Film Production: Enhancing Collaboration with MetaCrew. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641234.3671022>

1 INTRODUCTION

Virtual Reality (VR) is transforming industrial education, particularly in collaborative learning scenarios such as film training, where traditional film production education relies on expensive equipment and team collaboration [Van Wyk and De Villiers 2009; Wei et al. 2023]. This dependence often limits practical training and pushes instruction towards theory [Helzlsouer 2023]. To overcome these barriers, we designed a Virtual Reality Collaborative Learning (VRCL) system specifically tailored for film production – MetaCrew.

*Corresponding authors

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Posters '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0516-8/24/07

<https://doi.org/10.1145/3641234.3671022>

MetaCrew includes training roles for directors, cinematographers, and gaffers, and features a variety of lighting fixtures, cameras, interchangeable LED background screens, and numerous filming props. MetaCrew not only simulates a real environment to foster active learning and teamwork but also addresses the high costs and physical limitations of traditional methods. An important aspect of our VRCL system is enhancing social interaction, which is crucial for collaborative learning [Kasapakis and Dzardanova 2022]. Additionally, considering the challenge beginners face in mastering professional communication terminology quickly, we integrated a graphical user interface with visual symbols to aid clear communication between different roles in film production. Our research demonstrates the potential of VR in improving practical training in film production, highlighting our system’s contributions to immersive learning, enhanced communication, and inclusive, role-based education.

2 METHOD

We built the prototype system with Unity’s multiplayer networking for user communication and object synchronization. A user can create a room as a host, generating a join code for others to enter the same room. This system provided an affordable, immersive learning platform for film students and professionals by creating detailed Blender models of a film set, including elements like a soundstage, LED display, and camera, for a realistic VR simulation. Scene models were created using Blender.

We designed our film operation GUI through interviews with two film production experts, in conjunction with Mike’s visual symbol design guidelines [Zender and Mejía 2013], and the GUI design guidelines by Marcus et al. [Marcus 1995]. It also featured a graphical user interface with film-related symbols to improve communication and learning, enabling smoother interactions and task execution within the virtual set, as shown in Figure2.

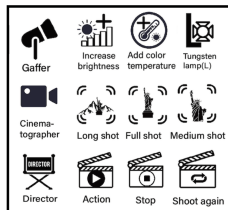


Figure 2: Film Operation GUI Design Partial Sample. MetaCrew Interaction Pipeline

As shown in Figure1(1)-(4), the pipeline provides a user-friendly and collaborative interface for LED virtual production training, where users, categorized as teachers or students, log in and enter a shared virtual environment with avatars and operational interfaces. It features two types of virtual light fixtures with size options that can be manipulated—added, deleted, or moved. Interactive controls for these fixtures allow adjustments in light intensity, illumination range, and RGB color settings, enhancing learning through real-time modifications. The system supports first-person and third-person views, facilitating immersive and flexible learning experiences as users engage with cinematic symbols for communication and collaboration. The code is available here: ¹.

¹https://drive.google.com/file/d/1jKBA7mGhXdrv8wnMIjR1g--Bt9g3IVqJ/view?usp=drive_link

3 EVALUATION

We recruited 24 participants, aged 20-40 years. These film production enthusiasts, all with good or corrected vision, were randomly divided into eight groups of three. Half used the standard MetaCrew and the other half used MetaCrew integrated with a film operation GUI. Among them, 10 of the standard group and 9 of the GUI group had prior VR experience. Participation was voluntary, approved by the university’s institutional review board, and each participant received \$10 USD upon completion.

Protocol. We investigated the GUI’s impact on VR-led film education through an experiment with groups using MetaCrew with and without a GUI. Participants engaged in a preliminary VR tutorial; GUI users also learned interface symbols. They answered social interaction questionnaires assessing copresence, interaction quality, and user experience on a 7-point scale.

Results. We evaluating the differences in social interactions within film production training, the Mann-Whitney U test was used to analyze the data from users of film operation systems with and without a graphical user interface (GUI). The study involved 24 participants. The results indicated that users with a GUI significantly outperformed those without a GUI in aspects of social interaction. Specifically, the group using the GUI scored an average of 5.00 (SD = 1.27) in team communication, which is 42.86% higher compared to the score of 3.50 (SD = 0.70) for the group without a GUI. This substantial difference highlights the critical role of the GUI in enhancing social interaction effectiveness in film production training environments.

4 CONCLUSION & FUTURE WORK

Our study showcases the educational benefits of the Virtual Reality-enabled film production Learning (VRCL) system in film education, particularly focusing on film production. We found that VR enhances understanding of film production workflows and the GUI design noticeably influences social interaction by facilitating clear communication and role-focused multitasking. Despite positive feedback on the VRCL system’s usability, participants noted challenges in avatar identification due to uniformity. Future improvements include customizable avatars and a gesture-controlled interface to enhance the learning experience and foster more intuitive interactions.

REFERENCES

- Volker Helzle. 2023. Immersive media productions involving light fields and virtual production LED walls. In *Immersive Video Technologies*. Elsevier, 575–589.
- Vlasios Kasapakis and Elena Dzardanova. 2022. Virtual reality learning environments: using high-fidelity avatars to enhance distance learning experience. *Interactive Learning Environments* (2022), 1–14.
- Aaron Marcus. 1995. Principles of effective visual communication for graphical user interface design. In *Readings in human-computer interaction*. Elsevier, 425–441.
- Etienne Van Wyk and Ruth De Villiers. 2009. Virtual reality training applications for the mining industry. In *Proceedings of the 6th international conference on computer graphics, virtual reality, visualisation and interaction in Africa*. 53–63.
- Zheng Wei, Xian Xu, Lik-Hang Lee, Wai Tong, Huamin Qu, and Pan Hui. 2023. Feeling Present! From Physical to Virtual Cinematography Lighting Education with Metashadow. In *Proceedings of the 31st ACM International Conference on Multimedia*. 1127–1136.
- Mike Zender and G Mauricio Mejía. 2013. Improving icon design: through focus on the role of individual symbols in the construction of meaning. *Visible Language* 47, 1 (2013), 66–89.